THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS V — MATHEMATICS · MODEL PAPER 09 ANSWER KEY

Answers and marking points | Maximum Marks: 80

UNDERSTANDING SHAPE AND SPACE — Marks 21

Q.	Marks	Answer / marking points
1	1	The side of each square is 1 cm. The perimeter of the shaded figure is **(b) 16 cm**
2	6	**(i)** Write the type of each angle shown: (a) ∠P (b) ∠Q (c) ∠R [3] (a) acute (less than 90°) (b) obtuse (just less than 180°) (c) reflex (more than 180°) **(ii)** (a) Through how many degrees does the hour hand turn from 12… [3] (a) 5 × 30 = **150°** (b) right angle (c) 360 ÷ 180 = **2**
3	6	**(i)** Name the solid that looks like: (a) a brick (b) a funnel (c)… [3] (a) cuboid (b) cone (c) square pyramid **(ii)** Use the table. (a) Which two solids have the same number of … [3] (a) **Triangular pyramid** (4, 4) and **square pyramid** (5, 5) (b) **Cube** (8) (c) 12 + 9 = **21**
4	8	The plan of Hari's house in Kota is given. **(a)** Find the perimeter of the Living Room. [2] 2 × (7 + 4) = **22 m** **(b)** Find the area of the Store. [2] 3 × 4 = **12 sq m** **(c)** In which direction (with respect to the Living Room) is the … [2] **East** (it is on the right side; north is at the top) **(d)** A carpet costing ₹ 85 per sq m is laid in the Bedroom. Find … [2] Area = 7 × 3 = 21 sq m; cost = 21 × 85 = **₹ 1,785**

NUMBER SYSTEM — Marks 9

Q.	Marks	Answer / marking points
5	1	In the number 6,08,347 the digit in the ten-thousands place is **(c) 0**
6	1	The smallest 6-digit number that can be made from the digits 7, 0, 4, 2, 9, 6 (each used o… **(a) 2,04,679**
7	1	705009 is read as **(d) seven lakh five thousand nine — 7,05,009**
8	6	**(i)** A potter of Kota made three pots weighing 1.25 kg, 2.5 kg an… [3] (a) 1.25 + 2.50 + 0.75 = **4.50 kg** (b) 2.50 − 0.75 = **1.75 kg** (c) less (4.5 kg < 5 kg) **(ii)** The line from 0 to 2 is divided into 8 equal parts. Write th… [3] A = 3/4 B = 5/4 = 1 1/4 C = 7/4 = 1 3/4

OPERATIONS — Marks 25

Q.	Marks	Answer / marking points
9	1	Which of these numbers is divisible by 9? **(c) 4,923 — 4 + 9 + 2 + 3 = 18, a multiple of 9**
10	3	A school in Jodhpur buys 28 benches at ₹ 650 each. The shopkeeper gives a discount of ₹ 1,… Cost of benches = 28 × 650 = ₹ 18,200 After discount = 18200 − 1200 = **₹ 17,000**
11	3	Divide 9,876 by 13. Write the quotient and the remainder, and check the answer by multipli… Quotient = **759**, remainder = **9**. Check: 13 × 759 = 9867; 9867 + 9 = 9876. ✓
12	6	**(i)** Ramlal sells 35 litres of milk every day at ₹ 52 per litre. … [3] (a) 35 × 52 = **₹ 1,820** (b) 1,820 × 30 = **₹ 54,600** (c) 54,600 − 18,000 = **₹ 36,600** **(ii)** Find using an easy method: (a) 52 × 98 (b) 36 × 25 (c) 4,000… [3] (a) 52 × 100 − 52 × 2 = 5200 − 104 = **5,096** (b) 36 × 100 ÷ 4 = 3600 ÷ 4 = **900** (c) 4,000 − 2,000 + 3 = **2,003**

Q.	Marks	Answer / marking points
13	6	There are 20 lockers numbered 1 to 20. Student 1 touches every locker, Student 2 touches e… **(i)** (a) Which students touch locker 12? (b) How many students to… [3] (a) **1, 2, 3, 4, 6, 12** (the factors of 12) (b) factors of 16 are 1, 2, 4, 8, 16, so **5** students (c) **1, 3, 9** **(ii)** (a) Which lockers are touched by exactly 2 students? (b) Whi… [3] (a) **2, 3, 5, 7, 11, 13, 17, 19** (prime numbers) (b) **6, 12, 18** (c) **12** (LCM of 4 and 6)
14	6	**(i)** Solve the riddles: (a) I am a multiple of both 6 and 8, and … [3] (a) **48** (b) **36** (3 + 6 = 9, 3 × 6 = 18; 63 is more than 40) (c) **64** (8 × 8) **(ii)** The floor of a room is covered by square tiles in 4 rows and… [3] (a) 4 × 7 = **28** (b) length = 7 × 50 = 350 cm = **3.5 m**; breadth = 4 × 50 = 200 cm = **2 m** (c) 3.5 × 2 = **7 sq m**

MEASUREMENT — Marks 17

Q.	Marks	Answer / marking points
15	3	Fill in the blanks: (a) 7 km 250 m = _____ m (b) 2 days = _____ hours (c) 3 L 50 mL … (a) 7250 (b) 48 (c) 3050
16	3	Write True or False: (a) 00:00 hours is midnight in the 24-hour clock. (b) 1 quintal is 10… (a) True (b) False (1 quintal = 100 kg) (c) True
17	3	Ramesh started his work at 6:50 am and worked for 7 hours 25 minutes. At what time did he … 6:50 am + 7 h = 1:50 pm; 1:50 pm + 25 min = 2:15 pm. He finished at **2:15 pm** (14:15).
18	8	The table shows the road distance (in km) between some cities. **(a)** A taxi goes from Jaipur to Ajmer and comes back to Jaipur. H… [2] 2 × 130 = **260 km** **(b)** Which city is nearest to Alwar? [2] **Bharatpur** (110 km) **(c)** Which city is farthest from Kota? [2] **Alwar** (400 km) **(d)** Going from Alwar to Jaipur through Bharatpur is how many km … [2] 110 + 180 = 290 km; 290 − 150 = **140 km**

DATA HANDLING AND PATTERNS — Marks 8

Q.	Marks	Answer / marking points
19	1	The product of two odd numbers is always **(a) an odd number — e.g. 3 × 5 = 15, 7 × 9 = 63**
20	1	Tiles are arranged in rectangles as shown. How many tiles will be needed for Fig. 6? **(d) 42 — 1×2, 2×3, 3×4, … → 6 × 7 = 42**
21	3	A pictograph shows the vegetables sold in a shop. Each dot (●) stands for 10 kg. **(a)** How many kg of vegetables were sold on Thursday? [1] 5 × 10 = 50 kg **(b)** On which day was the least sold? [1] Wednesday (3 baskets = 30 kg) **(c)** How many more kg were sold on Tuesday than on Wednesday? [1] 60 − 30 = 30 kg
22	3	Look at the pattern: 3, 4, 6, 9, 13, 18, ___, ___. (a) Write the number added at each step… (a) +1, +2, +3, +4, +5, … (the added number goes up by 1 each time) (b) 18 + 6 = 24, 24 + 7 = 31 (c) the 8th number is 31, the 9th is 31 + 8 = 39, so **8** is added

END OF ANSWER KEY — Total: 80 Marks